Constant-weight calibration and the cost of extending actual cores

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The first result replaces the variable-weight endpoint field by **any** constant-weight actual solution after a controlled conformal reduction. An exact nonnegative identity controls its amplitude, harmonic-curvature remainder and Maxwell residual, uniformly in the parameter. Keeping its two error scales separate improves the all-metric compactified gap from cubic to quadratic at the zero-Weyl end, and strengthens the upper energy wall on the same actual topological inputs. The subsequent calculations test the actual high-index cores against the missing global matching, and retain the curvature-action cancellation in the balanced first-order problem. Neither O10 nor the balanced optimal coefficient is settled here.

1. The weak endpoint can be studied at constant scalar curvature

Let the fixed Spin-c structure be solvable for every metric in the conformal class of a closed oriented four-metric g, with $p = (a_g^+)^2 > 0$. As in I07, put

$$S_2 = \int s^2, \quad P = \int |\operatorname{Ric}^\circ|^2, \quad w = \sqrt{6}|W^+|, \quad D_\beta = (1+\beta^{-1})S_2 + (\beta+1)\int w^2 - 72\pi^2 p,$$
$$r = V^{-1}\int |\operatorname{Ric}+g|^2, \qquad \kappa = r + D_\beta/V, \qquad \beta \geqslant 2. \tag{1.1}$$

The conformal class has negative Yamabe constant: a nonnegative constant scalar representative would contradict the positive actual monopole projection by the spinor Weitzenbock formula. Let $\widehat{g}$ be its scalar-minus-four representative. Hodge star on two-forms, p, and the integrated Weyl energies are conformally invariant. The exact identities proved in I41 Section 1 give

$$\delta = S_2 - \widehat{S}_2 \geqslant 0, \qquad \widehat{P} = P - \delta/12, \qquad \widehat{D}_\beta = D_\beta - (1+\beta^{-1})\delta \geqslant 0. \tag{1.2}$$

Thus $0 \leqslant \delta \leqslant D_\beta$. If $e = s + 4$, then $\int e^2 \leqslant 4rV$ and $S_2 = 16V - 8\int e + \int e^2$. Since $\widehat{S}_2 = 16\widehat{V}$,

$$1 - \sqrt{r} - \kappa/16 \leqslant \frac{\widehat{V}}{V} \leqslant 1 + \sqrt{r} + r/4. \tag{1.3}$$

In particular, when the left denominator is positive,

$$\boxed{\widehat{\kappa} = \frac{\widehat{P} + \widehat{D}_\beta}{\widehat{V}} \leqslant \frac{\kappa}{1 - \sqrt{\kappa} - \kappa/16}.} \tag{1.4}$$

An actual sequence with $\beta \to \infty$, $\kappa \to 0$ therefore exists if and only if such a sequence exists with $s = -4$. The same class, its square condition, and χ/A are retained; $\widehat{V}/V \to 1$. There is no pointwise Ricci or conformal-factor regularity claim. This is an endpoint consequence of the inherited exact conformal identities, not a new Yamabe existence theorem.

We will also need a comparison independent of D_β. Write $g = u^2\widehat{g}$, with $\Delta = d^*d$ on $\widehat{g}$. The scalar transformation gives $eu^2 = 6u^{-1}\Delta u + 4(u^2 - 1)$. Expand its square and integrate. The cross term is $48\int (u - u^{-1})\Delta u = 48\int (1 + u^{-2})|du|^2$. Subtract the exact formula for δ in I41 Section 1 to obtain

$$\boxed{\int_g (s_g+4)^2 = \delta + 16\int_{\widehat{g}} (u^2-1)^2 + 48\int_{\widehat{g}} |du|^2.} \tag{1.5}$$

In particular $\delta \leqslant 4rV$. Moreover,

$$16\widehat{V} = 16V - 8\int_g e + \left(\int_g e^2 - \delta\right) \geqslant 16V(1-\sqrt{r}), \qquad e = s_g + 4. \tag{1.6}$$

Thus $\widehat{V}/V \geqslant 1-\sqrt{r}$, whether or not D_β/V is small. This is an energy comparison, not a bound on the Hessian of the conformal factor in a uniformly controlled chart.

2. An exact constant-weight calibration for every finite parameter

Work now with $s = -4$. Take **any** actual solution

$$D_A\Phi = 0, \qquad F_A^+ = 2i\sigma(\Phi). \tag{2.1}$$

It exists by scaling an ordinary unperturbed solution: if its spinor is Ψ, take $\Phi = \Psi/\sqrt{2}$. No continuity, uniqueness, or comparison with the earlier variable-weight solutions is used. The maximum principle gives $0 \leqslant \rho = |\Phi|^2 \leqslant 2$. Set

$$\begin{gathered} \psi = 2\sqrt{2}\sigma(\Phi), \qquad \zeta = \langle \Phi, \nabla_A\Phi\rangle, \quad J = \|F_A^+ - (F_A^+)_H\|_2^2 \geqslant 0, \\ E_\lambda = \langle W^+\psi, \psi\rangle + w\rho^2/3 \geqslant 0, \\ D_0 = 2\int |d\rho|^2 + \int (4\rho|\nabla_A\Phi|^2 - |\nabla\psi|^2) + 2\int |d\psi|^2 + 2\int E_\lambda \geqslant 0. \end{gathered} \tag{2.2}$$

For $\beta \geqslant 2$, define

$$c = \frac{4(\beta+1)}{3\beta} \in [4/3, 2], \qquad z = w - 4/\beta, \qquad C_\beta(\rho) = \rho + c/2 - \frac{(\rho+c)^2}{2c(\beta+1)}. \tag{2.3}$$

Theorem (same-solution constant-weight identity).

$$\boxed{\begin{aligned} \frac{c}{18}D_\beta = \frac{1}{2}D_0 + cJ + \int (\rho-c)^2 C_\beta(\rho) \\ + \frac{c(\beta+1)}{18}\int \left[z - \frac{3(\rho^2-c^2)}{c(\beta+1)}\right]^2. \end{aligned}} \tag{2.4}$$

Every term on the right is nonnegative. In particular,

$$\boxed{D_0 \leqslant \frac{2}{9}D_\beta, \quad J \leqslant \frac{1}{18}D_\beta, \quad \int (\rho-c)^2 \leqslant \frac{1}{6}D_\beta.} \tag{2.5}$$

Proof. The actual spinor/form identities in I07, with weight exactly two, read

$$D_0 = \frac{2}{3}\int (4+w)\rho^2 - 2\int \rho^3. \tag{2.6}$$

Since $|F_A^+|^2 = \rho^2/2$, Hodge projection gives

$$9\int \rho^2 = 16(1+\beta^{-1})V + (\beta+1)\int w^2 - D_\beta + 18J. \tag{2.7}$$

Insert (2.7) into (2.6), use (2.3), and expand the cubic around c. Before completing the last square, the result is

$$\frac{1}{2}D_0 + cJ + \int (\rho-c)^2(\rho+c/2) + \frac{c(\beta+1)}{18}\int z^2 = \frac{c}{18}D_\beta + \frac{1}{3}\int z(\rho-c)(\rho+c). \tag{2.8}$$

Completing the z square gives (2.4). Further, $C_\beta(0) = 2/3$, and

$$C'_\beta(\rho) = \frac{\beta(4\beta+4-3\rho)}{4(\beta+1)^2} > 0 \quad (0 \leqslant \rho \leqslant 2).$$

Thus $C_\beta \geqslant 2/3$. Dropping terms gives (2.5), using $c \leqslant 2$. This proof never changes the actual solution. QED.

There is an equivalent full Weyl-square identity. Define the globally smooth polynomial model

$$\alpha = \frac{9\beta}{8(\beta+1)^2}, \qquad W_0 = \alpha(\rho^2 I/3 - \psi\otimes\psi).$$

Its two repeated eigenvalues are $\alpha\rho^2/3$. Direct expansion in the three-dimensional self-dual bundle gives

$$|W^+ - W_0|^2 = (w - 2\alpha\rho^2)^2/6 + 2\alpha E_\lambda.$$

The square in (2.4) is exactly $(w-2\alpha\rho^2)^2$. Using (2.9) below cancels the E_λ term and yields

$$\boxed{\frac{D_\beta}{18} = J + \frac{4}{c}\int |\zeta|^2 + \frac{\beta+1}{3}\int |W^+ - W_0|^2 + \frac{1}{c}\int C_\beta(\rho)(\rho-c)^2.} \tag{2.9a}$$

At $\beta = 2$, rescaling back to the ordinary spinor gives precisely the four-term identity in I47 Section 4.1. The extension here is the same-solution identity for the whole finite parameter range, not a new claim about the already established beta-two case. It requires only the actual solution (2.1) on the scalar-minus-four metric; conformal solvability was used to reach and supply that representative in Section 1.

2.1. A first-order Maxwell estimate without a spectral gap

The exact Hopf/Dirac identities used in I47 Section 4.1 are

$$d\rho = 2\,\mathrm{Re}\,\zeta, \quad |\nabla\psi|^2 = 4\rho|\nabla_A\Phi|^2 - 4|\,\mathrm{Im}\,\zeta|^2, \quad |d\psi|^2 = 2|\,\mathrm{Im}\,\zeta|^2.$$

Consequently

$$D_0 = 8\int |\zeta|^2 + 2\int E_\lambda. \tag{2.9}$$

The determinant Bianchi identity is $dF_A = 0$, and $F_A^+ = i\psi/\sqrt{2}$. In the project's Clifford convention, $\delta F_A = -2i\,\mathrm{Im}\,\zeta$. Its norm, independently of the sign convention, therefore obeys

$$\boxed{\|\delta F_A\|_2^2 = 4\|\,\mathrm{Im}\,\zeta\|_2^2 \leqslant D_0/2 \leqslant D_\beta/9.} \tag{2.10}$$

Let H_A be the harmonic representative of the same imaginary curvature class. The exact form $F_A - H_A$ has zero square integral, so its two chiral squared norms are equal. Hence

$$\boxed{\|F_A - H_A\|_2^2 = 2J \leqslant D_\beta/9, \qquad d(F_A - H_A) = 0, \qquad \|\delta(F_A - H_A)\|_2^2 \leqslant D_\beta/9.} \tag{2.11}$$

In fact the sum is bounded by the same constant: $\|F_A - H_A\|_2^2 + \|\delta(F_A - H_A)\|_2^2 \leqslant D_\beta/9$. Indeed (2.4) gives $\|\delta F_A\|_2^2 + (c/2)\|F_A - H_A\|_2^2 \leqslant cD_\beta/18$, and $c/2 \leqslant 1$.

This is a Hodge graph-norm bound. It is not a rough covariant $W^{1,2}$ bound on two-forms: the latter introduces the uncontrolled opposite Weyl curvature in the Weitzenbock formula. Nor does (2.11) bound $d\delta F_A$ in L^2. Spectrally, coefficients of size λ^{-1} at eigenvalue $\lambda \to \infty$ have vanishing graph norm but a nonvanishing Laplacian. Such a spectral illustration is not an actual counterexample to the coupled equations.

By (2.5), $\rho \to 4/3$ in normalized L^2 at squared error $O(D_\beta/V + \beta^{-2})$. With $\omega = (3\sqrt{2}/4)\psi$,

$$\|\omega\|_\infty \leqslant 3\sqrt{2}/2, \qquad \fint \left[(|\omega|^2 - 2)^2 + |d\omega|^2\right] \leqslant C(D_\beta/V + \beta^{-2}). \tag{2.12}$$

Thus the unrestricted actual endpoint can be attacked using a scalar-minus-four metric and a constant-weight solution with small Maxwell residual. All original supplier quantifiers remain in force; only the analytic field chosen for the proof has changed. No bound on ∇w or derivative of an auxiliary weight is needed.

2.2. Continuing the Maxwell equation to the critical source

The current has $\mathrm{Re}\,\zeta = d\rho/2$. Hence, in the same convention as (2.10), $d\zeta = -d\delta F_A/2$. Since $dF_A = 0$, the actual torsion curvature relation of I40 Section 3 becomes the global polynomial identity

$$\boxed{\rho^2\Delta_H F_A^- + \rho^3 F_A^- = -2q_-(t) + \mathscr{R},} \tag{2.13}$$

$$|\mathscr{R}| \leqslant C\left(\rho^3|\,\mathrm{Ric}^\circ\,| + \rho^{3/2}|\zeta||\nabla_A\Phi|\right).$$

Hodge Laplacian commutes with chirality. On $\{\rho > 0\}$ this is $(\Delta_H + \rho)F_A^- = -2q_-(t)/\rho^2 + \mathscr{R}/\rho^2$. The source has no singular norm bound there: the exact pointwise $|t|^2 \leqslant 2\rho^2|\nabla_A\Phi|^2$ and $|\zeta| \leqslant \sqrt{\rho}|\nabla_A\Phi|$ bound its magnitude by $C(\rho|\operatorname{Ric}^\circ| + |\nabla_A\Phi|^2)$. But its squared norm involves the critical fourth-power gradient energy. The available budget bounds that quantity, not its concentration by a small multiple of V. Division by ρ across the zero set is not needed or asserted in (2.13).

There is also a precise remaining term in a covariant-regularity attempt. For the real ASD form $a_- = -iF_A^-$,

$$\|\nabla a_-\|_2^2 = 2\|da_-\|_2^2 + \frac{4}{3}\|a_-\|_2^2 + 2\int\langle W^- a_-, a_-\rangle. \tag{2.14}$$

Even if $a_- = K + E$, with a bounded fitted model K, $\|E\|_2^2 = o(V)$, and $\|W^-\|_2^2 \leqslant C(B)V$, only the K,K and K,E terms are controlled by these norms. The remaining $\int\langle W^- E, E\rangle$ requires information not supplied by two separate L^2 bounds. Thus (2.13) is a concrete new way to use the full equations, not a regularity or zero-end-gap conclusion.

The natural signed pairing can be taken further. Along a hypothetical zero-end sequence assume $a^2 \geqslant T_+$, put $\eta = r + D_\beta/V + \beta^{-2} \leqslant 1$, and use normalized norms below. Set $\rho_* = 4/3$, $U_* = -4/3$, $m_* = 128/81$, $\Pi = \nabla_A\Phi$, $\tau = t_\Phi$, $m = |\tau|^2$, and $S = m_* + 2\rho^2|\Pi|^2$. The constant-weight identities give

$$\Delta\rho = -2|\Pi|^2 - (-4 + 2\rho)\rho/2, \qquad 0 \leqslant \nu := 2\rho^2|\Pi|^2 - m \leqslant 4\rho|\zeta|^2.$$

The estimates above and the inherited Hessian estimate imply $\|\rho - \rho_*\|_2 + \|\zeta\|_2 + \|d\rho\|_2 + \|\operatorname{Ric}^\circ\|_2 \leqslant C\sqrt{\eta}$, with Π, τ bounded in normalized L^4 and $\nabla_A\Pi, \nabla\tau$ bounded in normalized L^2. Thus $⨍ S = 2m_* + O(\sqrt{\eta})$, $⨍ S^2 \leqslant C$, and $⨍ \nu = O(\eta)$. The scalar test $2\rho^2/S$ has derivative

$$d(2\rho^2/S) = \frac{4m_*\rho}{S^2}d\rho - \frac{8\rho^4}{S^2}\operatorname{Re}\langle\Pi, \nabla_A\Pi\rangle.$$

It is bounded in normalized L^2, so this test gives $⨍ S^{-1} = (2m_*)^{-1} + O(\sqrt{\eta})$. The same nonnegative reciprocal defect as in I40, Section 8 then yields $⨍ m^2/S = m_*/2 + O(\sqrt{\eta})$ and $⨍ (m/S)^2 = 1/4 + O(\sqrt{\eta})$.

Write real ASD forms $f = -iF_A^-$, $Q = -iq_-(\tau)$, $R = -i\mathscr{R}$, and $B = \rho^3 Q/S$. Pairing the actual curvature equation with Q/S, whose weighted derivative test is bounded by I40 (8.10), gives $⨍\langle f, B\rangle = -m_*/8 + O(\sqrt{\eta})$ and $⨍ |B|^2 = \rho_*^6/32 + O(\sqrt{\eta})$. Chern–Weil supplies

$$⨍ |f|^2 + \frac{4\pi^2(a^2 - T_+)}{V} = ⨍ \rho^2/2 - 2⨍ |W^+|^2 - 2/3 + r/2 = 2/9 + O(\sqrt{\eta}).$$

Completing the square, exactly as in I41 Section 6, proves

$$K = -\frac{4m_*}{\rho_*^6}B, \qquad \|K\|_\infty \leqslant C, \qquad \|f - K\|_2^2 + \frac{4\pi^2(a^2 - T_+)}{V} \leqslant C\sqrt{\eta}. \tag{2.14a}$$

All denominators are bounded below by m_*; this transfers the bounded ASD fit to this same constant-weight solution without identifying it with an earlier weighted solution.

Let $\gamma = \operatorname{Im}\zeta$. Since $\delta f = -\gamma$, $df = *\gamma$, integration by parts in (2.13) gives the exact identity

$$\int \rho^3|f|^2 + 2\int\langle Q,f\rangle + 2\int \rho^2|\gamma|^2 = \int\langle R,f\rangle - 4\int \rho\langle\gamma,\iota_{\nabla\rho}f\rangle. \tag{2.14b}$$

The right side is $O(\eta^{1/4}V)$. Indeed R is bounded in normalized L^2 and has L^1 norm $O(\sqrt{\eta})$; the ASD form $\rho(d\rho\wedge\gamma)^-$ is similarly L^2-bounded and L^1-small. Pair their bounded-model parts with K and their remaining parts with $f-K$ using (2.14a). Testing instead against the harmonic $h=-i(H_A)^-$ gives the same formula with the test factor f replaced everywhere by h, without the nonnegative γ^2 term. Its error is also small by (2.11).

This closes the error in these signed tests, but their principal terms balance: $\oint\rho^3|f|^2\to 128/243$, whereas $2\oint\langle Q,f\rangle\to -128/243$. It is not a contradiction. Testing instead with $\Delta_H f$ leaves $4\int\langle\gamma,\delta Q\rangle$ and $\int\langle R,\Delta_H f\rangle$. The estimate $|\delta Q|\leqslant C|\tau||\nabla\tau|$ gives only a critical bounded contribution, not a vanishing one. This identifies the unpaid derivative coupling rather than assigning a false positive sign to (2.14b).

2.3. An anisotropic all-metric gap

The improved Weyl-fit coefficient in (2.9a) changes the parameter estimate; it should not be discarded by replacing every error with $r+D_\beta/V$.

Theorem. Under the actual conformal supplier of Section 1, with $b_a=(-a^2)_+/V$, there is a universal $c_0>0$ such that

$$\boxed{(1+b_a)\left(r+\frac{D_\beta}{\beta V}\right)\geqslant c_0\frac{(\beta-2)^2}{\beta^4},\qquad \beta\geqslant 2.} \tag{2.15}$$

If $a^2\geqslant T_+=2\chi+3\tau$, the factor $1+b_a$ can be removed. No pointwise curvature, Euler-density, diameter, noncollapse, Sobolev, or stationarity bound is assumed.

Proof on a scalar-minus-four metric. Put $t=1/\beta$, $d=D_\beta/V$, $\varepsilon=r+td$, and $\omega=\sqrt{2}\psi/c$, $v=|\omega|^2$. Then

$$W_0=t(vI/3-\omega\otimes\omega),\quad \oint|W^+-W_0|^2\leqslant Ctd,$$
$$\|\omega\|_\infty\leqslant C,\qquad \oint\left((v-2)^2+|dv|^2+|d\omega|^2\right)\leqslant Cd. \tag{2.16}$$

Here the endomorphism model has coefficient t, not $t/2$. All bounds follow from the same solution in (2.9a). Its coarse Hessian budget is $C(1+r+d+b_a)V$: the positive-spinor curvature uses only W^+, scalar curvature, Ricci and the determinant curvature; $\int|F_A^-|^2\leqslant 2V+4\pi^2(-a^2)_+$, while the scalar equation bounds $\int|\nabla_A\Phi|^2\leqslant CV$. More precisely, $\nabla_A^*\nabla_A\Phi=(1-\rho/2)\Phi$, and $\int|\nabla_A^*\nabla_A\Phi|^2\leqslant 8V/27$. Apply the bounded-section Hessian identity and interpolation exactly as in I07 Section 5; no derivative estimate for F_A is needed. No opposite Weyl or derivative-of-weight term enters this estimate.

Let $J_t=\operatorname{div}W(W_0)$, where $W(\cdot)$ is the usual passage from the self-dual endomorphism to its curvature tensor, and $X=\oint|J_t|^2$. The derivative of this polynomial has its explicit factor t, so $\oint|\nabla J_t|^2\leqslant Ct^2(1+r+d+b_a)$. The weak Bianchi self-test therefore gives

$$X \leqslant Ct\sqrt{(r+td)(1+r+d+b_a)}. \tag{2.17}$$

For completeness, the exact polynomial/Hodge contraction from round 21, Section 6, equation (17) on this form is

$$\begin{aligned} X = t^2 \oint \Big[&\tfrac{4}{3}v^2 - \tfrac{4}{3}tv^3 + 2v\langle (W^+ - W_0)\omega, \omega\rangle + 4v|d\omega|^2 - \tfrac{5}{6}|dv|^2 \\ &+ 4\langle \iota_{\nabla v}\omega, *d\omega\rangle \Big]. \end{aligned} \tag{2.18}$$

Its constant-amplitude part is $(16/3)t^2(1-2t)$. The amplitude, Weyl, and derivative errors in (2.18) are bounded, respectively, by $Ct^2\sqrt{d}$, $Ct^2\sqrt{td}$, and $Ct^2 d$. Thus

$$t^2(1-2t) \leqslant CX + Ct^2(\sqrt{d} + d + \sqrt{td}). \tag{2.19}$$

If $r, d \leqslant 1$, divide by $t > 0$, insert (2.17), and use $t\sqrt{d} \leqslant \sqrt{td}$, $td \leqslant \sqrt{td}$, and $t\sqrt{td} \leqslant \sqrt{td}$. This proves $t(1-2t) \leqslant C\sqrt{(1+b_a)(r+td)}$. If $r > 1$ or $d > 1$, its squared conclusion follows directly from $r + td \geqslant \min(1, t)$. This proves (2.15) on the representative.

Transfer to every input metric. For $r \leqslant 1/16$, (1.5) implies $\widehat{V}/V \geqslant 1 - \sqrt{r} \geqslant 3/4$, and

$$\hat{r} + t\widehat{d} = \frac{P + tD_\beta - (1/12 + t + t^2)\delta}{\widehat{V}} \leqslant 2(r + td), \qquad 1 + \widehat{b}_a \leqslant 2(1 + b_a).$$

Apply the just-proved estimate to $\widehat{g}$. For $r > 1/16$ the desired bound is immediate after reducing c_0. Finally, $a^2 \geqslant T_+$ and Chern–Weil imply $b_a \leqslant r/(8\pi^2)$; the small/large r split removes the factor. QED.

This proves a stronger estimate than the inherited compactified transfer, although its right side still tends to zero. For an exact normalized negative Einstein metric it gives $D_\beta/V \geqslant c_0(\beta-2)^2/\beta^3$ after square-cost removal. It does not give a positive lower bound independent of β.

2.4. The upper energy wall now has a quadratic endpoint margin

Use the compactified functional $P_t = tD_{1/t}$, extended at zero by $P_0 = 6\int |W^+|^2$. Equation (2.15) says

$$\boxed{(1 + b_a)(r + P_t/V) \geqslant c_0 t^2(1-2t)^2, \qquad 0 \leqslant t \leqslant 1/2.} \tag{2.20}$$

The prior exponent was three. This improvement is for integral Ricci error, not only the bounded-pointwise-Ricci setting of I45.

Suppose, as in the inherited upper-wall application, that every metric admits the specified actual selection with $p \geqslant A > 0$, $a^2 \geqslant T_+$, and $T_+ = A(1-h)$, $1/4 < h < 1/3$. Define

$$e = \frac{\int |\operatorname{Ric}|^2}{8\pi^2 A}, \quad G(h) = \frac{(4 + \sqrt{1-3h})^2}{9}, \quad t_* = \frac{1 - 2\sqrt{1-3h}}{8 + 2\sqrt{1-3h}}.$$

Then every smooth metric satisfies

$$\boxed{\mathscr{E}(g) + (G(h) - e)_+ \geqslant c_* t_*^2 (1 - 2t_*)^2.} \tag{2.21}$$

Proof. Retain the exact Chern–Weil identity from the compactified pencil, Section 3:

$$P_t = k(t)R + \ell(t)P + 12\pi^2(1+t)T_+ - 72\pi^2 tp,$$
$$k(t) = (1+t)(8t-1)/2, \qquad \ell(t) = 2(1+t)(1-2t), \quad R = \int |\operatorname{Ric}|^2.$$

At $t = t_* \in (0, 1/8)$, the topological terms cancel at $e = G(h)$, so $P_t \leqslant 8\pi^2 A|k(t)|(G(h) - e)_+ + \ell(t)P$. For normalized negative target error $r \leqslant 1$, the actual scalar budget gives $8\pi^2 A/V \leqslant 9$, and $P/V \leqslant r$. Combining with (2.20) gives (2.21) for the negative-target infimum by constant rescaling. Large residuals are immediate.

For clarity, passing to $\mathscr{E}$, the infimum over both signs, uses the inherited independent positive-target bound. If $I = \int s \leqslant 0$, that infimum over positive targets is four. Otherwise it is $\mathscr{E}_+ = 4 - I^2/(RV) \geqslant 4/e$, using $\int s_-^2 \geqslant 32\pi^2 A$. If $\mathscr{E}_+ < 1/2$, the other Chern–Weil budget gives $e(1 - \mathscr{E}_+) \leqslant 3\min(T_+, T_-)/A \leqslant 3$, so $e \leqslant 6$, contradicting $\mathscr{E}_+ \geqslant 2/3$. Thus $\mathscr{E}_+ \geqslant 1/2$, which also implies (2.21) after reducing its constant. QED.

As $h \downarrow 1/4$,

$$t_*^2(1 - 2t_*)^2 = \frac{4}{9}(h - 1/4)^2 + O((h - 1/4)^3).$$

This is an actual strengthening of the whole-metric upper wall on its existing supplied topological families. It is not a new manifold supply, not a proof that the quadratic rate is optimal, and not a nondegenerate zero-end estimate.

3. What high index costs on an amplitude-controlled local core

We next continue X07 rather than replace an actual core by a formal matrix. For fixed d, on the rescaled hyperbolic ball $x = ry$, take

$$\rho_0 = 4/3, \quad b = \rho_0 2^{(d-1)/2}, \quad t_0 = b(\bar{z}_1^d, \bar{z}_2^d), \qquad m_0 = |t_0|^2. \tag{3.1}$$

The antiholomorphic primitive and curved contraction of X07 give an exact solution (2.1), with $t(ry) = t_0(y) + O_d(r)$ in C^1, on taking r sufficiently small after d. This order of choices suffices for every diagonal assertion below; no uniform implicit-function radius is assumed.

Fix $0 < m_1 < m_2$. Work on the level core $\Omega_d = \{m_0 < m_2\}$ and its shell $T_d = \{m_1 < m_0 < m_2\}$, instead of the round collar whose amplitude grew rapidly with d. Set

$$K_d = \frac{\pi^2}{d^2 b^{4/d}} B(1/d, 1/d), \qquad B_d = B(2 - 1/d, 1/d).$$

For $k \geqslant 0$, elementary integration gives

$$\int_{\Omega_d} m_0^k = r^4 K_d \frac{m_2^{k+2/d}}{k+2/d}, \qquad K_d \sim \frac{\pi^2}{2d}. \tag{3.2}$$

In particular, volume is asymptotic to $\pi^2 r^4/4$, whereas positive moments of torsion mass are $\Theta(r^4/d)$. On the shell, writing $\Delta m^q = m_2^q - m_1^q$,

$$\int_{T_d} |\nabla t_0|^2 = 4\pi^2 r^2 b^{-2/d} \frac{d^2}{d+1} \Delta m^{1+1/d}, \tag{3.3}$$

$$\int_{T_d} |\nabla t_0|^4 = 4\pi^2 d^2 (B_d + 1) \Delta m^2, \tag{3.4}$$

$$\int_{T_d} |\langle t_0, \nabla t_0 \rangle|^2 |\nabla t_0|^2 = \frac{4\pi^2 d^2}{3} [(2 - 1/d) B_d + 1] \Delta m^3. \tag{3.5}$$

Verification. Put $u = b^2|z_1|^{2d}$, $v = b^2|z_2|^{2d}$. Then $d\mu = \pi^2 r^4 d^{-2} b^{-4/d} (uv)^{1/d-1} du\, dv$. The change $u = mx, v = m(1-x)$ proves (3.2). Use $|\nabla_y t_0|^2 = 2b^2 d^2 (|z_1|^{2d-2} + |z_2|^{2d-2})$ for (3.3)–(3.4), and $|\langle t_0, \nabla_y t_0 \rangle|^2 = 2b^4 d^2 (|z_1|^{4d-2} + |z_2|^{4d-2})$ for (3.5). The beta recurrence reduces the latter angular factor to $(2-1/d)B_d + 1$. Also $B_d = d - 1 + O(d^{-1})$, by the reflection formula for Gamma.

Thus quadratic derivative energy is $\Theta(r^2 d)$, while the two critical shell energies are $\Theta(d^3)$. The seed spinor satisfies $|\nabla \Phi_0|^2 = m_0/(2\rho_0^2)$ and $|\nabla^2 \Phi_0|^2 = |\nabla t_0|^2/(2\rho_0^2)$. For each fixed d, actual perturbations converge in every required finite norm as $r \to 0$; choose the diagonal so these relative errors also tend to zero. The regular positive mass levels persist.

The relative torsion index remains d^2; the determinant relative square is exactly zero by Stokes on the trivial line. The former uses the actual torsion boundary framing and is not an unframed determinant charge. On a boundary near $m_0 = m_* = 128/81$, mass is normal but outgoing derivatives are of size d/r. The whole-core mass mismatch is $\Theta(r^4)$, not small relative to its own volume. Both it and the negative determinant model mismatch have vanishing integrals after a sufficiently small insertion. A calibrated exterior is still absent.

3.1. What an extension must pay

Suppose an extension has a region U and compactly supported cutoff η, equal to one on the actual core, with

$$\operatorname{Vol} U \leqslant C r^4, \qquad |d\eta| \leqslant C/r, \qquad |t| \leqslant C \text{ on } U. \tag{3.6}$$

These are explicit extension hypotheses, not uniform geometric properties of the level shell: its thickness can be of order r/d, and its extrinsic curvature of order d/r. Amplitude control alone does not give the cutoff in (3.6). Put $E_D = \int_U |D_{3A}^- t|^2$ and $E_0 = \int_U (s^2 + |F_A^-|^2)$. The twisted Lichnerowicz identity applied to ηt gives

$$\boxed{c r^2 d \leqslant C E_D + C r^2 + C r^2 \sqrt{E_0}.} \tag{3.7}$$

Indeed the gradient energy on the core is bounded below by (3.3). The Dirac product rule costs at most $2E_D + Cr^2$. Its scalar and determinant-curvature potential, acting on a bounded section

supported in volume Cr^4, has absolute integral at most $Cr^2\sqrt{E_0}$. The opposite Weyl tensor does not occur in this spinorial potential. For sufficiently large d, (3.7) implies

$$E_D \geqslant cr^2 d \quad \text{or} \quad E_0 \geqslant cd^2. \tag{3.8}$$

For the same actual solution the torsion equation bounds the first cost by the integral of $C(|W^+|^2 + |\zeta|^2|\nabla_A\Phi|^2)$, under the bounded-amplitude convention. Thus matching requires an additional actual source, not merely a choice of boundary framing. Nevertheless, if $V \asymp d^2$, its required residual density is only $r^2 d/V \asymp r^2/d \to 0$. The alternative (3.8) therefore does not exclude a weak endpoint sequence.

4. Splitting index changes the critical cost

The d^3 cost of (3.5) is not universal for index d^2. Fix $0 < L < 1/2$ and use the separated seed

$$t_0 = b((\bar{z}_1/L)^d - 1, (\bar{z}_2/L)^d - 1). \tag{4.1}$$

Its antiholomorphic primitive in the same spinor construction is

$$g = \frac{bL}{2\rho_0}\left[\frac{(\bar{z}_1/L)^{d+1}}{d+1} - \frac{\bar{z}_1}{L} - \frac{(\bar{z}_2/L)^{d+1}}{d+1} + \frac{\bar{z}_2}{L}\right]. \tag{4.2}$$

There are $N = d^2$ simple zeros. For large b, each fixed mass component has inverse branches $\bar{z}_i = L\xi_i(1 + w_i/b)^{1/d}$, with $\xi_i^d = 1$. Their derivatives are similarities up to relative $O(b^{-1})$, uniformly in d. Each physical component has scale $\ell = rL/(bd)$ and index one. Take r sufficiently small after (d, b) for the actual contraction and its relative errors.

Summing over these components, the linear-map integrals give

$$\mathrm{Vol}\{m < m_2\} = (1 + o(1))\frac{\pi^2 m_2^2}{2} N\ell^4, \qquad \int_{m<m_2} |\nabla t|^2 = (1 + o(1))2\pi^2 m_2^2 N\ell^2, \tag{4.3}$$

$$\int_{m_1<m<m_2} |\nabla t|^4 = (1 + o(1))8\pi^2 N\Delta m^2, \tag{4.4}$$

$$\int_{m_1<m<m_2} |\langle t, \nabla t\rangle|^2 |\nabla t|^2 = (1 + o(1))\frac{8\pi^2}{3} N\Delta m^3, \tag{4.5}$$

$$\int_{m_1<m<m_2} |\nabla(t/|t|)|^4 = (1 + o(1))9\pi^2 N \log(m_2/m_1). \tag{4.6}$$

For example $t = x/\ell$ has $|\nabla t|^2 = 4/\ell^2$ and $|\nabla(t/|t|)|^2 = 3/|x|^2$; radial integration proves the constants. The actual perturbation preserves these asymptotics, the simple indices, and zero determinant relative charge on every component.

Here $N\ell^2 = r^2L^2/b^2$ and $N\ell^4 = r^4L^4/(b^4d^2)$. Thus the subcritical costs vanish, while the critical cost is only linear in total index. These are costs on the selected components, not an estimate on all the space between them.

This also identifies a qualification in I50's frontier interpretation. Its lower bound in terms of $(H/V)^{-1/3}$, $H = \sum 1/n_e$, forces divergent critical density **only if** $H/V \to 0$. A split collection of $N \asymp V$ trivial-character interfaces has $n_e = 1$ and $H/V \asymp 1$, with bounded positive critical density as in (4.6). I50's determinant atoms do not charge these interfaces. The local pattern fits this alternative; it is not a closed zero-end construction.

4.1. A closed extension needs a nonparallel exterior

There is a global test of the proposed matching which does not require extracting any cores. For the same weight-two solution on a closed scalar-minus-four metric, the integrated spinor equation is

$$K := \int |\nabla_A \Phi|^2 = \int k(\rho), \qquad k(\rho) = \rho - \rho^2/2.$$

Since $|k'| \leqslant 1$ on $[0,2]$, (2.5) gives

$$\boxed{\frac{D_\beta}{V} \geqslant 6\left|\frac{K}{V} - \frac{4}{9}(1 - \beta^{-1} - 2\beta^{-2})\right|^2.} \tag{4.7}$$

Thus a zero-end sequence must have $K/V \to 4/9$. For every fixed split seed the whole shrinking construction ball, including the regions between its selected components, has $K = O_{d,b}(r^4)$ and curvature energy $O_{d,b}(r^4)$; its Hessian integral is $O_{d,b}(r^2)$. This follows by differentiating the smooth rescaled seed and choosing the actual perturbation radius after (d,b), as in X07. A diagonal can make these costs negligible. Any closed extension with the required volume normalization must therefore put $(4/9 + o(1))V$ kinetic energy in the exterior. A proposed nearly parallel exterior with $K/V \to 0$ instead has $\liminf D_\beta/V \geqslant 32/27$.

This matching assertion assumes the closed scalar-minus-four extension actually retains the displayed local solution. A conformal reduction of some other proposed extension need not preserve its hyperbolic cores.

The harmonic data give another necessary exterior requirement. Under the class-square condition, the topological saturation already proved in I41, Section 6 gives $(a^2 - T_+)/V \to 0$ on the hypothetical zero-end sequence. This statement concerns the class and metric and does not identify its earlier weighted solution with the present one. Chern–Weil gives $T_+/V \to 1/(6\pi^2)$, while (2.5) gives $\int |F_A^+|^2/V \to 8/9$. Hence

$$\frac{\|H_A^+\|_2^2}{V} \to \frac{8}{9}, \qquad \frac{\|H_A^-\|_2^2}{V} \to \frac{2}{9}, \tag{4.8}$$

using the same-class harmonic remainder (2.11). The shrinking construction ball cannot carry these volume-scale harmonic energies: its actual curvature energy and the global harmonic remainder are both negligible at that scale. The exterior must supply them. It is not enough to glue the displayed local zero sets into a metrically convenient region.

These are necessary matching tests, not an exclusion of every exterior. A nonparallel bulk with $K/V \to 4/9$ and the harmonic data (4.8) is not contradicted by these identities. Its construction or exclusion, including the original same-class conformal solvability, remains open.

5. The balanced first-order target must retain curvature cancellation

Return to the **ordinary**, unscaled solution used in I50 Section 4 on the balanced family. This section does not substitute the weight-two field of Section 2 into ordinary-solution identities. Retain $\rho \leqslant 4$, χ, q_χ, B_o, J, M_χ, and P_o from that section. Put

$$f = 1 - \chi, \quad u = \Phi/\sqrt{\rho}, \quad \mathbb{F} = F^{S^+}, \quad C_\pm = \mathbb{F}^\pm u,$$
$$X_\pm = \int f|C_\pm|^2, \qquad Y_\pm = \int f \operatorname{Re}\langle C_\pm, F_A^\pm u\rangle. \tag{5.1}$$

The weighted expressions extend by zero near the spinor zero set. In a unitary frame whose first vector is u, write $\mathbb{F}^- = \left(\begin{smallmatrix} i\alpha & -\bar{b} \\ b & i\delta \end{smallmatrix}\right)$. Then

$$|\operatorname{Ric}^\circ|^2 = 2|\alpha - \delta|^2 + 8|b|^2, \qquad |F_A^-|^2 = |\alpha + \delta|^2.$$

Expanding both sides proves

$$\boxed{P_o - 2B_o = 8(X_- - Y_-), \qquad |Y_-| \leqslant \sqrt{X_- B_o}.} \tag{5.2}$$

Consequently the existing spatial balance becomes exactly

$$P - 8\pi^2 Ah = 8\left[X_- - Y_- - 4\pi^2 q_\chi + \pi^2(p - A) + J/4 + M_\chi/4\right]. \tag{5.3}$$

A sufficient joint target for coefficient four is $[Y_- - X_- + 4\pi^2 q_\chi]_+ = o(Ah)$. In particular, $X_- = o(Ah)$ and $(q_\chi)_+ = o(Ah)$ would suffice, since $B_o = O(Ah)$. Unlike demanding $B_o = o(Ah)$, this permits a nonzero ASD determinant field annihilating the actual spinor.

There is no automatic new sign: the same rank-two Chern identity gives

$$4\pi^2 q_\chi = (X_- - Y_-) - (X_+ - Y_+), \qquad Y_- - X_- + 4\pi^2 q_\chi = Y_+ - X_+. \tag{5.4}$$

Thus the coupled criterion is an exact reformulation of the surviving signed defect, not an independently proved positive estimate.

5.1. The same ordinary solution only has an order-Ah bound

Let $\Pi = \nabla_A \Phi$, $K = \int |\Pi|^2$, and $D = D_2$. Use the polynomial model from the ordinary remainder with $s_b = -4$, $\operatorname{Ric}_b^\circ = 0$, $W_b^+ = W_0$, $F_{A,b}^+ = i\sigma(\Phi)$, $F_{A,b}^- = 0$. Its curvature-action coefficients factor as

$$\rho^2/24 + 1/3 - \rho/4 = (\rho - 2)(\rho - 4)/24, \quad \rho^2/48 - 1/3 = (\rho - 4)(\rho + 4)/48.$$

The actual remainder bounds therefore imply $\int |\mathbb{F}_b \Phi|^2 \leqslant CD$ and $\int |\mathbb{F} - \mathbb{F}_b|^2 \leqslant C(P + B + D)$. Apply the exact Hessian identity of I40 Section 4.1 to this ordinary solution. Its bounded background terms cost CK; its error terms cost $C\sqrt{P + B + D}\|\Pi\|_4^2$. The bounded-section interpolation $\|\Pi\|_4^2 \leqslant 8\|\nabla_A^2 \Phi\|_2$, followed by Young, gives

$$\int (|\nabla_A^2 \Phi|^2 + |\nabla_A \Phi|^4) \leqslant C(K + P + B + D). \tag{5.5}$$

On the balanced $\mathscr{E} = O(h)$ sequences, the right side and X_- are $O(Ah)$, not $o(Ah)$.

The remaining signed interaction can be written without an abstract regularity assumption. Set $\mathscr{J}_3 = \mathrm{Re}\langle \Pi \wedge \mathbb{F}\Phi \rangle$, $\mathscr{C}_4 = \mathrm{Re}\langle \Pi \wedge \mathbb{F} \wedge \Pi \rangle$, with conjugation in the first Hermitian factor. Covariant differentiation and Bianchi give

$$d\mathscr{J}_3 = (|\mathbb{F}^+\Phi|^2 - |\mathbb{F}^-\Phi|^2) d\mu - \mathscr{C}_4.$$

Integrate against $w_0 = f/\rho$, extended smoothly by zero:

$$X_- = \int w_0 |\mathbb{F}^+\Phi|^2 - \int w_0 \mathscr{C}_4 + \int dw_0 \wedge \mathscr{J}_3. \tag{5.6}$$

The first term is $O(D)$. After subtracting the model, the critical interaction is bounded by $C\|\mathbb{F} - \mathbb{F}_b\|_2 \|\Pi\|_4^2 = O(Ah)$. The cutoff current is likewise $O(Ah)$. Taking absolute values therefore does not prove the required smaller signed remainder.

For an exact negative cscK canonical ordinary solution, $\rho = 4$, $\nabla_A \Phi = 0$, $C_\pm = 0$, and $\chi = q_\chi = 0$. Equation (5.2) correctly gives $P = 2B$, even when $B \neq 0$. This is why its cancellation must be retained. No transfer of this observation to a whole balanced handle-and-blowup recovery sequence is asserted without matching the actual solutions.

6. The remaining global implication

The endpoint problem now has an exact constant-scalar, constant-weight form with small determinant Hodge graph error. Its unresolved source is not an auxiliary weight derivative. It is the critical coupling of the torsion derivative, the actual curvature, and signed transition flux.

High-index cores cannot be excluded by their listed subcritical costs; their controlled extensions satisfy the explicit alternative (3.8). Splitting the zeros avoids treating the monomial's superlinear cost as universal. Neither model supplies the closed exterior, the global square condition, or the same-class conformal supplier required by O10.

For the balanced problem, (5.3)–(5.6) locate the order-Ah signed term without discarding the parallel-core cancellation. Its little-o estimate or a lower-coefficient actual recovery construction is still required. These are the two continuing global tests; the new identities do not replace them by a completed defect-decomposition or realization theorem.